

PROBLEM SET 1

Question 1

Consider a household living for two periods and maximizing:

$$\max_{c_1, c_2} u(c_1) + \beta E[u(c_2)] \quad (1)$$

$$s.t. \quad c_1 = Y_1 - B_1 - S_1 \quad (2)$$

$$c_2 = Y_2 + R_f B_1 + \tilde{R} S_1 \quad (3)$$

Where Y_1 denotes the deterministic income and c_t the consumption for period $t \in [1, 2]$. B_1 denotes the bond holdings and S_1 the stock holdings chosen by the household during period 1. R_f is the riskless gross return on bonds. The gross return on stocks, $\tilde{R}$, has a known distribution. The household has a CRRA utility function of the form $u(c) = \frac{c^{1-\rho}}{1-\rho}$.

- (a) Assume that we do not impose separate non-negativity constraints on bonds and stocks, but we require instead that the household is not a net borrower: $B_1 + S_1 \geq 0$. Set up the Lagrangean and state the complementary slackness conditions for the borrowing constraint.
- (b) Interpret the complementary slackness conditions.
- (c) State first order conditions for B_1 and S_1 .
- (d) Would we be able to determine B_1 and S_1 separately if $\tilde{R}$ was deterministic?
- (e) Show using *reductio ad absurdum* that stock non-participation cannot be optimal.
Hint: You are allowed to assume the presence of a strictly positive equity premium $E[\tilde{R}] > R_f$ for your proof.

Question 2

Discuss the following statements. Justify your claims or arguments with reference to relevant models and empirical findings wherever possible.

- (a) *In the standard portfolio choice model, the optimal risky portfolio share declines with normalized cash on hand. The intuitive explanation for this is that richer households do not need to generate high returns on their saving and can therefore afford to take less risk.*
- (b) *In a theoretical model of portfolio choice, greater levels of trust or greater optimism about the stock market have similar effects on the stock market participation decision. Using survey questions, we may be able to separate the two.*

Question 3

A researcher studies the role of education for the household decision to hold stocks.

She wants to estimate a model that relates a dummy for stockholding (the dependent variable) to education (main explanatory variable of interest), controlling for age, household size, gender, income, wealth and country fixed effects. The results of the corresponding Probit estimation are reported in Columns 1 and 2 in Table 1. The unit of analysis is a household. The main variable of interest is *education*, that corresponds to the years of education of the household head (being the person in charge of household finances).

Based on Columns 1 and 2, discuss the sign, magnitude and statistical significance of the estimated relationship of education and stockownership.

Table 1: Dependent variable: Stockholding participation				
VARIABLES	(1) coefficients	(2) MEM	(3) coefficients	(4) MEM
education	0.047*** (0.013)	0.003*** (0.001)	0.084*** (0.005)	0.014*** (0.001)
age	-0.001 (0.006)	-0.000 (0.000)	-0.007*** (0.002)	-0.001*** (0.000)
Household size	-0.032 (0.053)	-0.002 (0.003)	0.078*** (0.018)	0.013*** (0.003)
Male gender	0.320*** (0.101)	0.021*** (0.007)	0.310*** (0.036)	0.051*** (0.006)
INCOME controls	YES	YES	NO	NO
WEALTH controls	YES	YES	NO	NO
COUNTRY fixed effects	YES	YES	YES	YES
Observations	3,851	3,851	18,185	18,185

MEM – marginal effects at the mean. Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1